

Problem 2

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Problem 1 Find all real numbers x which satisfy each of the following equations.

(a) $\log_x(25) = 2$.

Explanation. Recall that, by definition of the inverse to an exponential function,

$$\log_b(x) = y \iff x = b^y.$$

Using this relationship we have $\log_x(25) = 2 \iff x^2 = 25$. Therefore

$$\begin{aligned} x^2 = 25 &\implies x = \pm 5, \\ &\implies x = 5 \quad (\text{base of log is always } > 0). \end{aligned}$$

Therefore $x = 5$ is the only solution to $\log_x(25) = 2$.

(b) $7^x = 15$

Explanation. Similar to the previous problem,

$$\log_b(x) = y \iff x = b^y.$$

Using this relationship we have $7^x = 15 \iff x = \log_7(15)$. Therefore $x = \log_7(15)$ is the only solution to $7^x = 15$.

(c) $\ln(x) + 1 = 0$.

Explanation. Similar to the previous two problems we'll use the relationship

$$\ln(x) = y \iff x = e^y.$$

Before applying this relationship we perform a bit of algebra first:

$$\ln(x) + 1 = 0 \implies \ln(x) = -1.$$

Now we have $\ln(x) = -1 \iff x = e^{-1}$. Therefore $x = e^{-1} (= 1/e)$ is the only solution to $\ln(x) + 1 = 0$.